Day 067 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Write One-Step Equations from Context

Standards: 6.EE.B.6, 6.EE.B.7

Learning target: I can model a situation with a one-step equation and solve it.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 11, 20

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01 / Learn

Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation.

Worked example 1

A collection grows by 6 cards to 42 cards. Let x be the starting count. Write and solve an equation.

1. Define the unknown quantity.

2. Translate the relationship into an equation before calculating.

3. Subtract 6 from both sides

4. 36 + 6 = 42.

5. State the result in the form requested: x + 6 = 42; x = 36 cards.

Answer: x + 6 = 42; x = 36 cards

Worked example 2

5 identical ropes have total length 15/2 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

1. Define the unknown quantity.

2. Total = number of ropes × one length.

3. Divide both sides by 5

4. 15/2 ÷ 5 = 3/2.

5. Check 5 × 3/2 = 15/2.

Answer: 5x = 15/2; x = 3/2 meters

Day 067 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A collection grows by 7 cards to 35 cards. Let x be the starting count. Write and solve an equation.

02 / Constructed response

A collection grows by 2 cards to 6 cards. Let x be the starting count. Write and solve an equation.

03 / Constructed response

A collection grows by 7 cards to 49 cards. Let x be the starting count. Write and solve an equation.

04 / Constructed response

7 identical lengths of rope total 14 m. Let x be the length of one piece. Write and solve an equation, then check.

05 / Constructed response

2 identical lengths of rope total 16 m. Let x be the length of one piece. Write and solve an equation, then check.

06 / Constructed response

5 identical lengths of rope total 35 m. Let x be the length of one piece. Write and solve an equation, then check.

07 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

08 / Equation entry and two-part response

2 identical ropes have total length 3 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

09 / Equation entry and two-part response

3 identical ropes have total length 21/2 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

10 / Constructed response

6 identical lengths of rope total 18 m. Let x be the length of one piece. Write and solve an equation, then check.

Day 067 | Unit 05 | Answer key

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1. x + 7 = 35; x = 28 cards

Subtract 7 from both sides; 28 + 7 = 35.

6.EE.B.6, 6.EE.B.7 | Constructed response

2. x + 2 = 6; x = 4 cards

Subtract 2 from both sides; 4 + 2 = 6.

6.EE.B.6, 6.EE.B.7 | Constructed response

3. x + 7 = 49; x = 42 cards

Subtract 7 from both sides; 42 + 7 = 49.

6.EE.B.6, 6.EE.B.7 | Constructed response

4. 7x = 14; x = 2 m

Divide the total by 7: 14 ÷ 7 = 2.

6.EE.B.6, 6.EE.B.7 | Constructed response

5. 2x = 16; x = 8 m

Divide the total by 2: 16 ÷ 2 = 8.

6.EE.B.6, 6.EE.B.7 | Constructed response

6. 5x = 35; x = 7 m

Divide the total by 5: 35 ÷ 5 = 7.

6.EE.B.6, 6.EE.B.7 | Constructed response

7. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

8. 2x = 3; x = 3/2 meters

Total = number of ropes × one length. Divide both sides by 2; 3 ÷ 2 = 3/2. Check 2 × 3/2 = 3.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

9. 3x = 21/2; x = 7/2 meters

Total = number of ropes × one length. Divide both sides by 3; 21/2 ÷ 3 = 7/2. Check 3 × 7/2 = 21/2.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

10. 6x = 18; x = 3 m

Divide the total by 6: 18 ÷ 6 = 3.

6.EE.B.6, 6.EE.B.7 | Constructed response